

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2263

Westin-Elgin 138 kV
March 23, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2263. The Interconnection Customer proposes a 162.1 MW battery energy storage facility interconnecting at the Westin-Elgin 138 kV substation in the Kentucky area. The request was received on December 26, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 662 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,048,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bozeman S Substation-Anson N Substation 138 kV	Cooper-Lee County Solid Waste Energy 345 kV	108.6	353	8.8
W Lebanon-Oakdale Power Sta 138 kV	New amp-Hubbrdstown 138 kV	108.3	243	7.2
Mcclellan Ar-Temple Station 243 138 kV	Westin-Elgin-Mcclellan Ar 138 kV line	105.3	372	44.4
Temple Sta 243-Westin-Elgin 138 kV Line	Westin-Elgin-Mcclellan Ar 138 kV	104.2	408	44.4
Delano-Frontier 138 kV	Harvest Power Orlando-Frontier 138 kV	103.4	237	23.3
Brunswick Cellulose-Bozeman S 138 kV	Universal Studios Tap Substation-Cooper 138 kV line	102.7	366	6.3
Stelco-Mammoth 138 kV Line	Pine Bluff Watson Chapel-Carraizo 1201 345 kV	102.3	199	7.3
Garden Villas-Davie 138 kV L	Union-Desoto Pleasant Run Substation 138 kV	102.0	180	5.5
Louisiana 1-New Camp 138 kV	W Leanon-Oakdale Power Station Substation 138 kV line	101.8	243	6.4
Mcclellan Ar-Spatsum Lornex 138 kV	Yuba City Sta A-Westin-Elgin 138 kV	101.7	764	81.4
Yuba City Sta A-Frontier 138 kV	Spatsum Lornex Substation-Mcclellan Ar 138 kV Line	101.6	740	77.1
Westin-Elgin-Yuba City Sta A 138 kV	Spatsum Lornex-Mcclellan Ar 138 kV	101.3	755	77.1
Kettle-Delano 138 kV line	Frontier-Harvest Power Orlando 138 kV	100.9	227	23.3
Mcclellan Ar-Westin-Elgin 138 kV line	Yuba City Sta A-Westin-Elgin 138 kV Line	100.8	601	59.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
New Camp Substation-Hubbardstown 138 kV	W Lebanon-Oakdale Power Sta 138 kV Line	100.1	249	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bozeman S Substation-Anson N Substation 138 kV	Reconductor line	\$10,753,000	7% (\$757,000)
W Lebanon-Oakdale Power Sta 138 kV	Reconductor line	\$4,900,000	4% (\$194,000)
Temple Station 243-McClellan Ar 138 kV	Reconductor line	\$1,288,000	18% (\$234,000)
Temple Sta 243-Westin-Elgin 138 kV Line	Reconductor line	\$1,391,000	18% (\$252,000)
Frontier-Delano 138 kV line	Reconductor line	\$1,873,000	22% (\$407,000)
Brunswick Cellulose-Bozeman South 138 kV	Reconductor line	\$1,835,000	8% (\$149,000)
Stelco Substation-Mammoth 138 kV Line	Reconductor line	\$11,354,000	7% (\$784,000)
Garden Villas-Davie 138 kV L	Reconductor line	\$3,359,000	5% (\$174,000)
Louisiana 1-New Camp 138 kV	Reconductor line	\$3,673,000	4% (\$151,000)
Spatsum Lornex-McClellan Ar Substation 138 kV Line	Reconductor line	\$3,717,000	22% (\$811,000)
Yuba City Sta A-Frontier 138 kV Line	Reconductor line	\$3,442,000	24% (\$810,000)
Westin-Elgin-Yuba City Sta A 138 kV	Reconductor line	\$5,267,000	24% (\$1,239,000)
Delano-Kettle 138 kV	Reconductor line	\$2,650,000	22% (\$575,000)
McClellan Ar-Westin-Elgin 138 kV line	Reconductor line	\$4,562,000	22% (\$996,000)
New Camp Substation-Hubbardstown 138 kV	Reconductor line	\$12,526,000	4% (\$515,000)
	Total allocated to project		\$8,048,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.